Set Exercises, Functions, and Elementary Problem-Solving

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§1.1 Sets as predicates, Boolean operations, and counting

If a set A has n elements, then it has

$$2^n$$

subsets. One way to see this is to use the characteristic function

$$\chi_S(x) = \begin{cases} 1, & x \in S, \\ 0, & x \notin S. \end{cases}$$

Choosing a subset $S \subseteq A$ is the same as choosing a function

$$\chi_S : A \to \{0, 1\}.$$

There are 2 choices for each element of A, hence 2^n choices in total.

This is a useful conceptual improvement over simply memorizing 2^n. A subset is a yes/no decision for every element. The subsequent algebraic and measurable-set material follows standard Boolean-algebra and probability presentations, in particular Kruckman's *Model Theory* notes on Stone duality and Scott Sheffield's MIT 18.175 lecture on sigma-algebras.[1]

De Morgan's laws are:

$$C_U(A \cap B) = C_U A \cup C_U B,$$
$$C_U(A \cup B) = C_U A \cap C_U B.$$

It also recalls inclusion–exclusion:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The exercises ask for set descriptions such as:

$$\{2n : 1 \leq n \leq 5\},$$

$$\{x : x \leq 5\},$$

and graphs such as

$$\{(x, y) : y = x^2 - x - 1\}.$$

This is a good reminder that sets can be finite lists, solution sets of inequalities, or geometric loci.

A subset $A \subseteq U$ is equivalently a predicate or a characteristic function

$$\mathbf{1}_A : U \to \{0, 1\}.$$

Set operations become operations on 0-1 functions:

$$\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B, \qquad \mathbf{1}_{A^c} = 1 - \mathbf{1}_A,$$

[1] A. Kruckman's logic lecture notes; S. Sheffield's MIT probability lectures.

and

$$\mathbf{1}_{A\cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A\mathbf{1}_B.$$

For finite U, cardinality is summation:

$$|A| = \sum_{x\in U} \mathbf{1}_A(x).$$

This converts set identities into ordinary algebraic identities.

§1.2 Two-set exercises and the atoms of information

One exercise gives

$$U = A \cup B = \{1,2,3,4\}, \qquad A \cap C_U B = \{1,2\},$$

and asks for B. Since $A \setminus B = \{1,2\}$, the remaining elements $3,4$ must lie in B, so

$$B = \{3,4\}.$$

Another exercise has

$$A = \{x : 1 < x < 7\}, \qquad B = \{x : x^2 - 4x - 5 \le 0\}.$$

Since

$$x^2 - 4x - 5 = (x-5)(x+1),$$

we get

$$B = [-1,5].$$

Therefore

$$A \cap C_{\mathbb{R}} B = (5,7).$$

Venn diagrams force the essential distinction between "inside A, inside B, inside both, inside neither."

One problem describes a survey of 100 students reading classical novels. If 80 read either X or Y, 60 read X, and 40 read both, then the number reading Y is found by

$$|X \cup Y| = |X| + |Y| - |X \cap Y|.$$

Thus

$$80 = 60 + |Y| - 40,$$

so

$$|Y| = 60.$$

Again, the important point is not the story, but the relation among union, intersection, and individual counts.

A Venn diagram is a visual partition of a universe into atoms determined by a finite family of sets. For two sets, the atoms are

$$A \cap B, \quad A \cap B^c, \quad A^c \cap B, \quad A^c \cap B^c.$$

For n sets, there are at most 2^n atoms. Every Boolean expression in the sets is a union of some of these atoms.

This gives a systematic method for elementary set identities: reduce both sides to the same atoms. In logic, the same method is truth tables. In probability, the atoms are elementary events. In database theory, they are the records satisfying combinations of predicates.

Thus the elementary operation of shading regions is already the algebra of information partitions. A set is a yes/no question about elements; a finite family of sets is a finite collection of yes/no questions; the atoms are the complete answer patterns.

§1.3 What the ambient number system allows

The note includes exercises with sets such as

$$A = \{a : a = x^2 - y^2,\ x, y \in \mathbb{Z}\},$$

and

$$B = \{a : a = x^2 - y^2,\ x, y \in \mathbb{Q}\}.$$

Since

$$x^2 - y^2 = (x + y)(x - y),$$

every difference of squares is a product of two numbers of the same parity in the integer case. This can be made into a complete characterization. An integer a is a difference of two integer squares exactly when $a \not\equiv 2 \pmod 4$. Indeed, $x+y$ and $x-y$ have the same parity. If both are odd, their product is odd; if both are even, the product is divisible by 4. Conversely, every odd integer $2k+1$ is $(k+1)^2 - k^2$, and every multiple $4k$ is $(k+1)^2 - (k-1)^2$. Over $\mathbb{Q}$ every rational q is a difference of squares, since

$$q = \left(\frac{q+1}{2}\right)^2 - \left(\frac{q-1}{2}\right)^2.$$

The representation problem is therefore controlled exactly by the ambient ring.

For rationals, the factorization is more flexible, and quotients of such expressions can often remain in B. The page is training the habit of factoring before trying random examples.

A floor-function example studies values of

$$y = \lfloor x \rfloor + \lfloor 2x \rfloor + \lfloor 3x \rfloor, \qquad 1 \le y \le 100.$$

The natural move is to write

$$x = k + t, \qquad k \in \mathbb{Z}, \quad 0 \le t < 1.$$

Then the three floor terms change only when t crosses multiples of $1/2$ or $1/3$. Hence the possible values in one period come from the intervals determined by

$$0, \quad \frac{1}{3}, \quad \frac{1}{2}, \quad \frac{2}{3}, \quad 1.$$

Indeed, $\lfloor x \rfloor + \lfloor 2x \rfloor + \lfloor 3x \rfloor = 6k + r(t)$, where $r(t) = 0, 1, 2, 3$ on the consecutive intervals $[0, \frac{1}{3})$, $[\frac{1}{3}, \frac{1}{2})$, $[\frac{1}{2}, \frac{2}{3})$, and $[\frac{2}{3}, 1)$. Hence the attainable integers are exactly those congruent to $0, 1, 2, 3 \pmod 6$; among the integers 1 through 100, exactly 67 are attained. This finishes the piecewise calculation rather than leaving it as a method hint.

§1.4 Functions as maps and the contravariance of preimages

A function is not merely a formula. It consists of a domain, codomain, and assignment rule. The same formula can define different functions if the domain or codomain changes. This is why image, preimage, injectivity, and surjectivity must always be stated relative to a specified domain and codomain.

For domains, ranges, monotonicity, and extrema of functions, typical tools include:

(i) rewriting a square root or logarithm condition as a domain restriction;

(ii) using monotonicity to locate maxima and minima;

(iii) drawing a rough graph to understand intervals;

(iv) using substitution when a composite expression repeats.

For an iterated piecewise map, tracing successive images across intervals reveals possible periodic orbits. The transition pattern is a small instance of symbolic dynamics: a finite graph of interval-to-interval transitions encodes which itineraries the iterates can follow.

For a function $f : X \to Y$, the preimage operation

$$f^{-1} : \mathcal{P}(Y) \to \mathcal{P}(X)$$

preserves Boolean operations:

$$f^{-1}(B \cup C) = f^{-1}(B) \cup f^{-1}(C),$$
$$f^{-1}(B \cap C) = f^{-1}(B) \cap f^{-1}(C), \qquad f^{-1}(Y \setminus B) = X \setminus f^{-1}(B).$$

The distinction can be checked on a two-point counterexample. Let $f : \{a, b\} \to \{0\}$ be constant and take $A = \{a\}$, $B = \{b\}$. Then $f(A \cap B) = \varnothing$, whereas $f(A) \cap f(B) = \{0\}$. In contrast, inverse images commute with arbitrary unions and intersections by logical substitution: $x \in f^{-1}(B \cap C)$ iff $f(x) \in B$ and $f(x) \in C$. The stability of inverse images is the reason topology defines continuity through preimages of open sets.

§1.5 Sigma-algebras make countable questions measurable

In probability and measure theory, one does not always allow every subset of a universe. A sigma-algebra $\mathcal{A}$ on U is a collection of subsets satisfying

$$U \in \mathcal{A}, \qquad A \in \mathcal{A} \Rightarrow A^c \in \mathcal{A}, \qquad A_1, A_2, \ldots \in \mathcal{A} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}.$$

This is a Boolean algebra closed under countable unions. The elements of $\mathcal{A}$ are the measurable sets or events.

The elementary set laws still hold, but the important new question is: which subsets are allowed to be measured? Probability begins by assigning a number $P(A)$ to each $A \in \mathcal{A}$ such that

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n)$$

for disjoint events. Thus basic set operations become the grammar of probability.

§1.6 Topology from Boolean Algebra

Stone duality says that Boolean algebras can be represented as algebras of clopen subsets of topological spaces. Given a Boolean algebra B, its Stone space consists of ultrafilters on B. Each element $b \in B$ determines a clopen set

$$\widehat{b} = \{\mathfrak{u} : b \in \mathfrak{u}\}.$$

For a finite Boolean algebra $\mathcal{P}(U)$, the ultrafilters are precisely the families of subsets containing a fixed element $u \in U$; the resulting Stone space is just the finite discrete set U. In general the Stone representation theorem identifies a Boolean algebra with the clopen subsets of its compact, Hausdorff, totally disconnected Stone space. This concrete finite case gives a precise bridge between truth-table atoms and the more abstract ultrafilter construction.